

Anti-Theft Smart Locker System using ESP32

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Abstract—This project presents an Anti-Theft Smart Locker System using ESP32. The system enhances security through password authentication, vibration-based intrusion detection, and IoT-based alert mechanisms. A hidden compartment mechanism protects valuables during unauthorized access attempts. The system provides fast response, reliable detection, and real-time alerting.

Index Terms—ESP32, Embedded System, IoT, Smart Locker, Security System

I. INTRODUCTION

In contemporary living, security of personal property is of the utmost importance. Many individuals keep jewelry, money, documents and electronic devices in lockers; assuming complete security. Current locker models only have basic mechanisms for locking and do not provide real-time monitoring of the use of the locker.

As a result of the rise of Internet of Things (IoT) based systems and embedded technologies, lockers can be transformed into intelligent security devices. An intelligent locker can provide a means to authenticate users, detect theft attempts and trigger alarms as well as send notifications through email, SMS and/or push notifications.

In this project, an **ESP32** will be utilized as a central controller which will handle the authentication process, detect theft attempts and trigger alarms for a single user. The **ESP32 microcontroller** system is ideal for these applications due to its integrated Wi-Fi capability, processing speed and multiple sensor interfaces.

A **microcontroller** is a compact programmable integrated circuit used to control electronic systems. An **embedded system** is a dedicated computing system designed for a specific function.

Security is a major concern in modern environments. Traditional lockers lack intelligent monitoring and fail to provide real-time alerts. This project introduces a smart locker system using **ESP32** that integrates authentication, sensing, and communication technologies.

The system detects unauthorized access and secures valuables using a hidden compartment, making it more advanced than conventional lockers.

II. LITERATURE REVIEW

A **literature review** is a study of existing research related to a topic.

Early era lockers began with mechanical locks that were operated with physical keys. Although these types of lockers

have been a simplistic method of providing security to people's property, the mechanical lock system can be easily duplicated and forced opened.

Later on, the advent of digitized locker systems (i.e., key-pads) allowed people to access their lockers using passwords. They did, however, provide little or no user convenience in respect of remote access or notifications when their locker had been tampered with.

The newest forms of security systems utilize a **microcontroller** platform, like Arduino or **ESP32**, equipped with various sensors. Many have alarms to notify users of a potential breach of security; however, the majority do not provide any type of automated means of locking the locker once someone has gained access to it.

The LockSecure project integrates user authentication, detection of intrusion, alarm generation, user notification **via IoT**, and a method of concealing the locking mechanism, into one comprehensive solution.

III. PROBLEM STATEMENT

Older type lockers have a number of limitations, such as:
No real-time alerts

- Can be easily compromised or damaged; keys can be lost or duplicated
- No monitoring system for unauthorized access attempts
- No alert for owner of unauthorized access attempts
- Slow to respond when an incident of theft occurs

A new solution needs to be developed that automatically detects any suspicious activity, alerts the user immediately, and secures their possessions in a more effective manner.

IV. OBJECTIVES

Older style lockers have several disadvantages including:

- They lack real time alerts.
- They are easily accessible and can be tampered with, and the key to them can easily be lost or duplicated.
- There is no system in place to monitor and alert the owner of unauthorized access attempts.
- They are slow to respond when there has been an incident of theft.

Therefore a new solution needs to be created that will automatically notify users when there are any suspicious activities associated with their locker and help secure their belongings more effectively.

V. SYSTEM DESIGN

A. Methodology of Model Development

Methodology is a structured approach used to design and implement a system.

Steps:

- 1) Password input via keypad
- 2) Validation by ESP32
- 3) Servo control for locking
- 4) Vibration detection
- 5) Alert triggering
- 6) Telegram notification

B. Hardware Module Design

The system consists of ESP32, keypad, servo motors, LCD, vibration sensor, and buzzer. Each module performs a specific function.

C. Block Diagram

A **block diagram** is a graphical representation of system components.

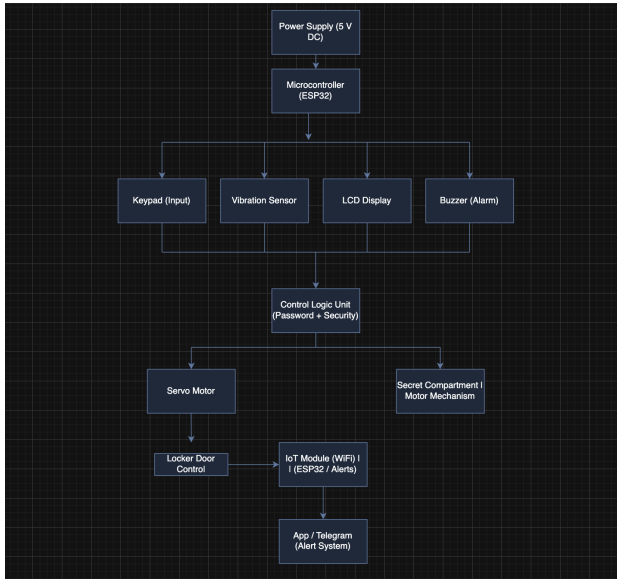


Fig. 1. System Circuit / Block Diagram

VI. HARDWARE DESCRIPTION

A. ESP32

ESP32 is a highly capable microcontroller that integrates Wi-Fi as well as Bluetooth (BLE). This device provides many GPIOs (input/output pins), low power consumption, as well as fast performance and can also be used to control all the sensors attached to an output device within a single system.

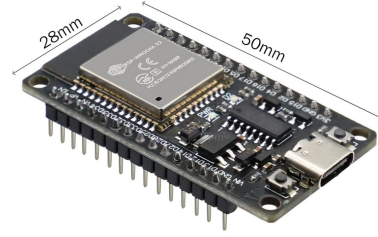


Fig. 2. ESP32 Module

B. Keypad

A **keypad** is an input device used for entering passwords.



Fig. 3. 4x4 Keypad

C. Servo Motor

The **Servo Motor** is a high-precision rotary actuator designed for accurate control of angular position, velocity, and acceleration. Unlike standard DC motors that rotate continuously, a servo is engineered for applications where specific positioning is critical.



Fig. 4. Servo Motor

D. Vibration Sensor

SW-420 Vibration Sensor is a module that can detect vibrations or shocks on a surface. It can be used for various purposes, such as detecting door knocks, machine malfunctions, car collisions, or alarm systems. It operates from 3.3 V to 5 V. The module has three peripherals, two LEDs, one for the power status and the other for the sensor output. In addition, there is a potentiometer that can be further used to control the threshold point of the vibration.



Fig. 5. Vibration Sensor (SW-420)

E. LCD Display

A **16x2 LCD** is a basic, affordable, and widely used Liquid Crystal Display module capable of showing 32 alphanumeric characters across two rows, with 16 characters per row

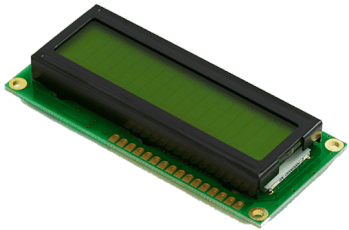


Fig. 6. 16x2 LCD

F. I2C Module

An **I2C Module** enables communication with microcontrollers like Arduino or ESP32 using only two data pins (SDA and SCL) instead of the 10+ wires required for parallel mode.

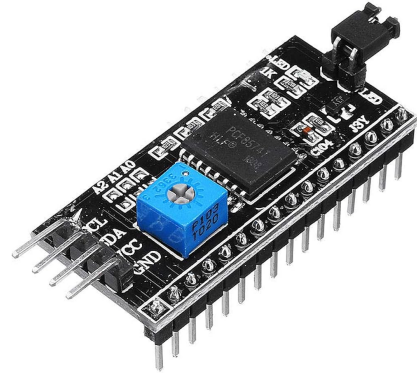


Fig. 7. I2C Module

G. Jumper Wires

Jumper wires are flexible electrical cables used in electronics. They have connectors at both ends and are often made of thin copper wire with insulation. They're handy for creating temporary connections between components on breadboards or circuit boards during testing or prototyping.

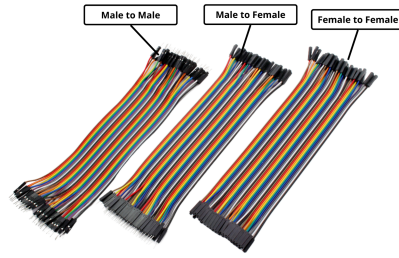


Fig. 8. Jumper Wires (M2M, F2F, M2F)

VII. HARDWARE WORKING

The system works by combining input, processing, and output.

When a password is entered, ESP32 validates it. If correct, the servo unlocks the locker. If incorrect, attempts increase and an alert is triggered after 3 failures.

If vibration is detected, the buzzer activates and the secret compartment secures valuables.

VIII. SOFTWARE OVERVIEW

An **IDE (Integrated Development Environment)** is software used to develop programs.

The system uses Arduino IDE and libraries such as Keypad, ESP32Servo, and LiquidCrystal_I2C.



Fig. 9. Arduino IDE

IX. SOFTWARE IMPLEMENTATION

An **algorithm** is a step-by-step procedure to solve a problem. **IF-ELSE** is a conditional statement used for decision-making.

The system performs:

- Password validation
- Servo control
- Alert generation
- IoT communication via HTTP

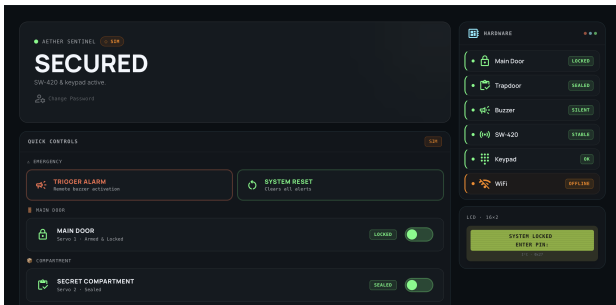


Fig. 10. Web Control UI

X. EXECUTION OF MODEL

The system was tested under different scenarios:

- Correct password → locker opens
- Wrong password → alert triggered
- Vibration → alarm + hidden compartment

XI. ADVANTAGES

- Improved security over normal lockers
- Real-time alerts to owner
- Password-based access control
- Intrusion detection using sensor
- Automatic locking mechanism
- Low cost and compact design
- Easy to install and maintain
- Suitable for IoT applications

XII. APPLICATION

- Home security
- Office lockers
- Banks
- Smart storage systems

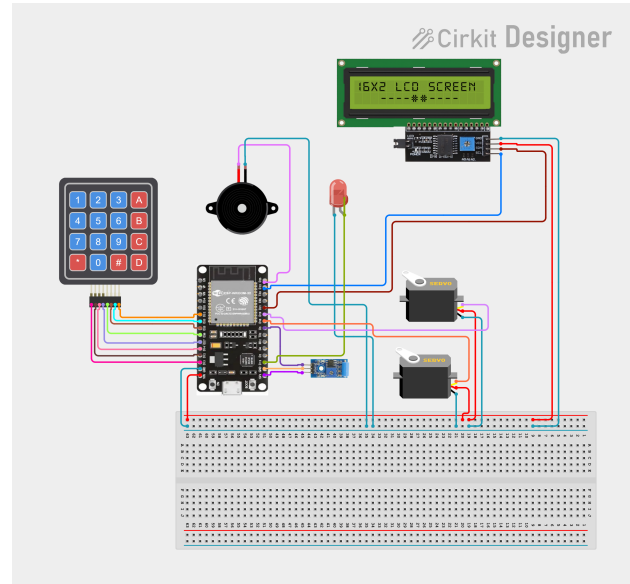


Fig. 11. Circuit

XIII. FUTURE SCOPE

- **Biometric Security** (authentication using fingerprint/face): Future improvements may include integrating AI-based face detection and fingerprint authentication systems to enhance security and automation. These features can provide multi-level user verification, allowing only authorized individuals to access the system. By combining computer vision, biometric authentication, and embedded technology, the project can become smarter, more reliable, and suitable for advanced real-world security applications.
- **Mobile App Control** (remote system management via smartphone): Enable users to control the locker, change password, and monitor status through a mobile application.
- **Cloud Integration** (online data storage and access): Use cloud platforms such as Firebase Realtime Database to store logs and manage system data.
- **Advanced Sensors** (high-precision motion detection): Integrate sensors like MPU6050 for improved and reliable tamper detection.
- **Real-Time Alerts** (instant notification system): Enhance alert mechanisms using internet-based communication for immediate user notification.
- **Power Backup System** (alternative energy source): Incorporate battery backup to ensure uninterrupted operation during power failures.

- **Multi-Locker System** (scalable locker management): Extend the system to manage multiple lockers in institutions such as banks or hostels.
- **OTA Updates** (over-the-air firmware updates): Allow remote software updates without requiring physical access to the system.

XIV. CONCLUSION

The Anti-Theft Smart Locker System with Secret Compartment and IoT Alerts successfully demonstrates a secure and intelligent approach to protecting valuables. The system integrates password-based authentication, vibration-based tamper detection, and an automated secret compartment mechanism to enhance security.

The implementation using a microcontroller (Arduino/ESP32) ensures efficient control of input, processing, and output components such as keypad, sensors, servo motor, and buzzer. The addition of IoT-based alerting enables real-time monitoring and notification, improving responsiveness to potential threats.

Overall, the project achieves its objective of providing a low-cost, reliable, and automated security solution. It also serves as a strong foundation for further enhancements such as biometric authentication, cloud-based control, and advanced intrusion detection systems.

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